Project 3: Ion Drive Rocket Analysis

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The purpose of this project is to

Nomenclature

T	=	temperature $[K]$
P	=	pressure $[Pa]$
ρ	=	density $\left[kg/m^3\right]$
T_t	=	total temperature $[K]$
P_t	=	total pressure $[Pa]$
u	=	velocity $[m/s]$
M	=	mach number
A	=	area $\left[m^2\right]$
F	=	thrust $[N]$
R	=	gas constant $[J/kgK]$
C_p	=	constant pressure specific heat $[J/kgK]$
γ	=	specific heat ratio
$\dot{m}$	=	mass flow rate $[kg/s]$
I_{sp}	=	specific impulse $[s]$
g_0	=	standard gravitational acceleration $\left[9.81m/s^2\right]$

I. Introduction

The purpose of this project temperature and pressure information is shown in table **??**. This data was used when running the solver at each altitude.

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II. Methodology

A. Rocket Analysis

The following equation can be used to calculate the propellant mass required to perform a maneuver.

$$\frac{\Delta v}{C} = \ln\left(\frac{m_0}{m_0 - m_p}\right) - \int_{t_0}^{t_f} \frac{D}{Cm} dt - \int_{t_0}^{t_f} \frac{g\cos\theta}{C} dt \tag{1}$$

Because the drag term and the gravity can be ignored, this equation can be simplified to the following.

$$m_p = m_0\left(1 - e^{\frac{-\Delta v}{C}}\right) \tag{2}$$

C in this equation is the effective exhaust velocity which can be found by using the following.

$$C = I_{sp} g_0 = 34335.0\ m/s \tag{3}$$

The required propellant mass for each stage of the flight is:

$$m_{p_a} = 51134.167\ kg$$

$$m_{p_d} = 51134.167\ kg$$

This gives a total propellant mass of:

$$m_p = 69584.546\ kg$$

Using the assumption that after the deceleration all the propellant is spent (the only remaining mass is that of the engine and payload), knowledge of the initial mass and payload mass can be used to calculate the engine mass. The engine mass m_e can be calculated using the equation below. Here m_0 is the initial mass, m_p is the propellant mass, and m_L is the payload mass.

$$m_e = m_0 - \left(m_p + m_L\right) = 5415.454\ kg \tag{4}$$

This engine mass is then used in subsequent ion drive calculations.

B. Ion Drive Analysis

$$I_{sp} = \frac{1}{g_0}\sqrt{2\frac{q}{m}\left(\phi_A - \phi_C\right)} \rightarrow \left(\phi_A - \phi_C\right) = V \tag{5}$$

$$e \tag{6}$$

C. Brayton Cycle Analysis

$$\pi_t = \frac{1}{\pi_c} \tag{7}$$

$$\frac{T_4}{T_3} = 1 - \eta_t \left(1 - \pi_t^{\frac{\gamma-1}{\gamma}}\right) \tag{8}$$

$$\frac{T_2}{T_a} = 1 + \frac{1}{\eta_c} \left(\pi_c^{\frac{\gamma-1}{\gamma}} - 1\right) \tag{9}$$

$$\eta = 1 - \frac{T_4 - T_a}{T_3 - T_2} \tag{10}$$

$$\frac{A_{rad}}{W_{out}} = \frac{\left(\frac{1}{\eta} - 1\right)}{3(T_4 - T_a)\sigma\epsilon} \left(\frac{1}{T_a^3} - \frac{1}{T_4^3}\right) \tag{11}$$

III. Results

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IV. Conclusion

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V. Appendix A